Kinematics

Motion of Projectile

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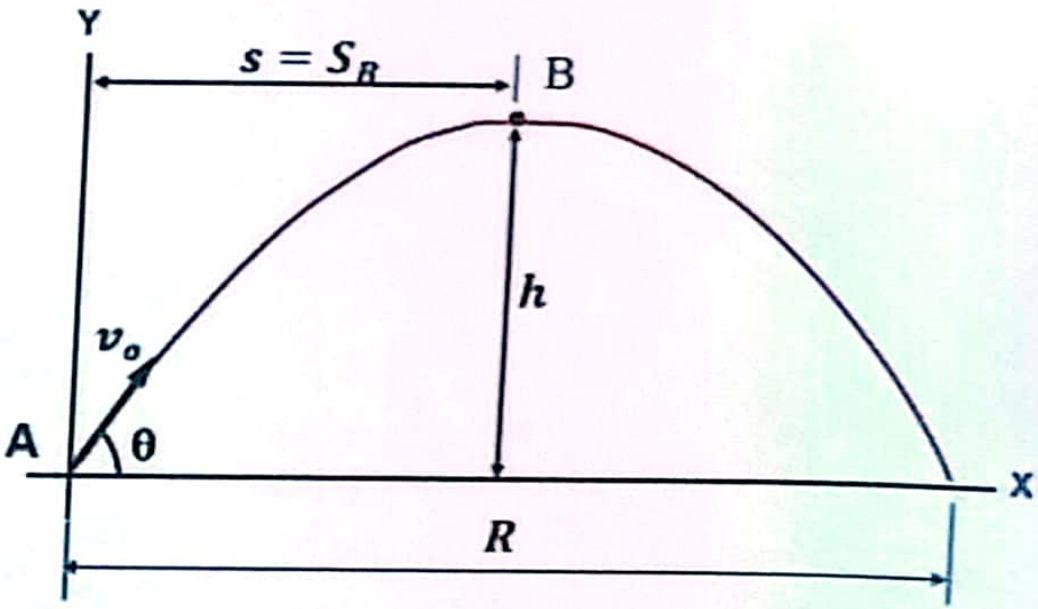
Sample Problem

Initial velocity v_o at an angle θ

Acceleration components are:

$$a_x = 0$$

$$a_y = -g$$

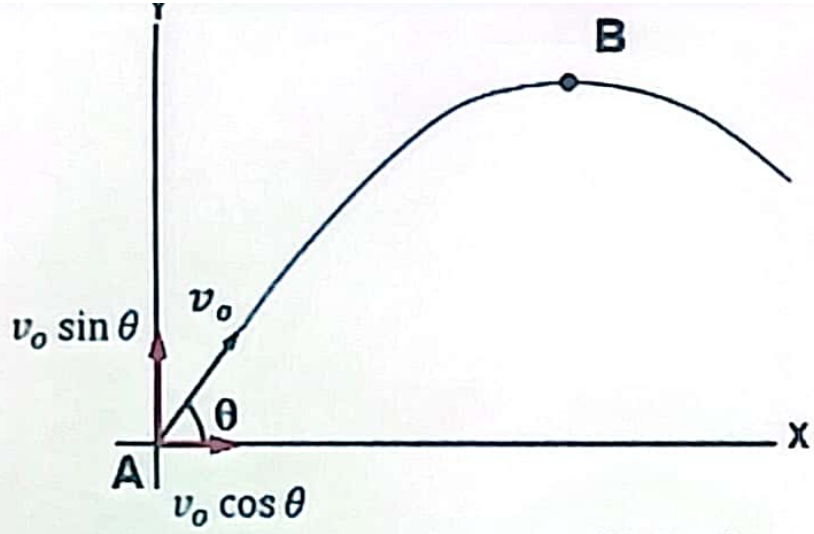


What is required?

Expressions for:

- maximum height reached, h
- horizontal distance at the point of maximum height, S_B
- time to reach maximum height, t_B
- range of motion, R
- equation of the path of motion

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Consider X-Motion:

$$a_x = 0$$

Velocity: $v_x = v_0 \cos\theta$

Position: $x = v_x t$

$$= (v_0 \cos\theta)\, t$$

Consider Y-Motion:

$$a_y = -g$$

Velocity: $v_y = v_{o_y} + a_y(t)$ ($-g$)

$$\Rightarrow v_y = v_o \sin\theta - gt$$

Position:

$$y = v_{o_y}(t) + \frac{1}{2} a_y (t)^2 \quad (-g)$$

$$\Rightarrow y = v_{o_y} t - \frac{1}{2} g t^2$$

$$\Rightarrow y = v_o t \sin\theta - \frac{gt^2}{2}$$

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Now we consider the projectile at point B:

What do we know? $v_y = 0$

$$v_y = 0 = v_o \sin\theta - g t_B \quad \Rightarrow \quad t_B = \frac{v_o \sin\theta}{g}$$

when $t = t_B$, then $y = h$: (use y)

$$y = v_o t \sin\theta - \frac{g t^2}{2}$$

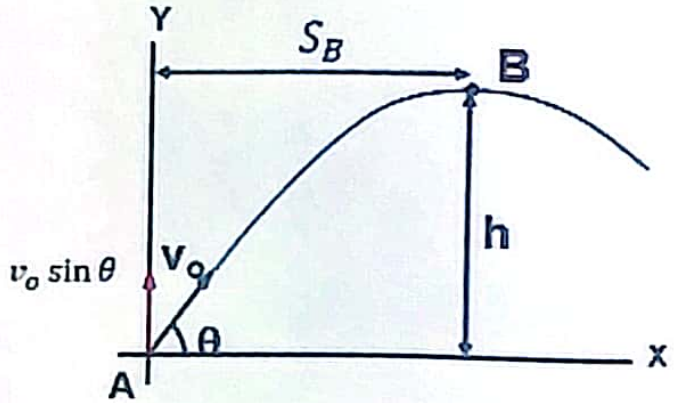
$$\therefore h = v_o\left(\frac{v_o \sin\theta}{g}\right)\sin\theta - \frac{1}{2} g \left(\frac{v_o \sin\theta}{g}\right)^2 \quad \Rightarrow \quad h = \frac{{v_o}^2 \sin^2\theta}{2g}$$

when $t = t_B$, then $x = S_B$:

$$x = v_o t \cos\theta$$

$$x = v_o\left(\frac{v_o \sin\theta}{g}\right)\cos\theta \quad \Rightarrow \quad S_B = \frac{{v_o}^2 \sin\theta \cos\theta}{g}$$

$$S_B = \frac{{v_o}^2 \sin 2\theta}{2g}$$



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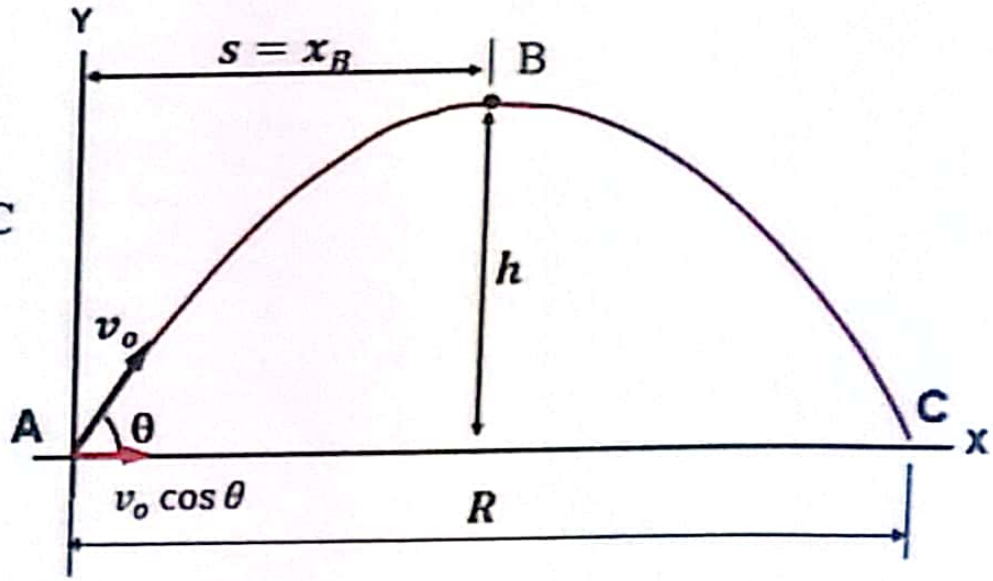
Range of motion, R

The time taken to reach B =Time taken from B to C

$$t_C = 2t_B = 2\left(\frac{v_o \sin\theta}{g}\right)$$

$$\because\ v_x = v_0 \cos\theta = \text{constant}$$

$$\therefore\ R = v_x t_C = (v_0 \cos\theta)\left(2\frac{v_o \sin\theta}{g}\right) \Rightarrow R = \frac{(v_0)^2 \sin 2\theta}{g}$$



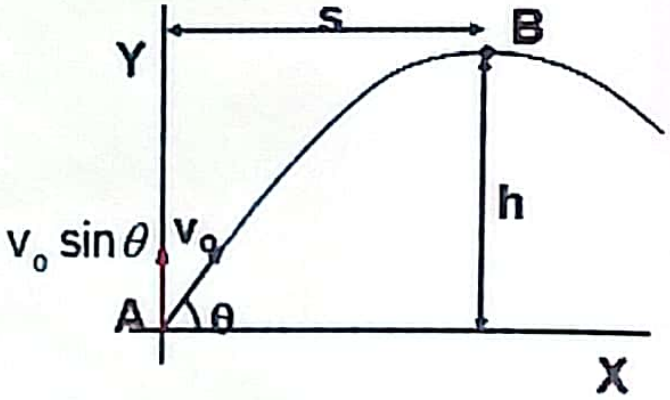
Equation of Path:

How? By eliminating t from x(t) and y(t):

$$x = (v_0 \cos\theta)t \quad \Longrightarrow \quad t = x/v_0 \cos\theta$$

Substituting this into $y(t) = v_o t \sin\theta - \frac{gt^2}{2}$

$$\Rightarrow \quad \boxed{y(x) = x\tan\theta - \frac{gx^2}{2{v_o}^2}\sec^2\theta}$$



$$\sec\theta = \frac{1}{\cos\theta}$$

EXAMPLE

A long-jumper leaves the ground at an angle of 20° to the horizontal and at a speed of 11 m/s.

(a) How far does he jump?

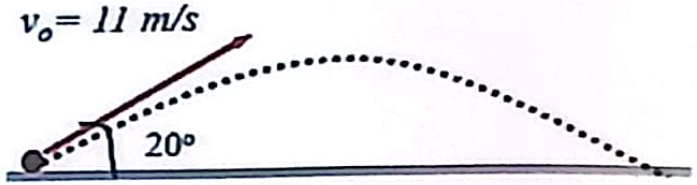
(b) What is the maximum height reached?



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(a) *The range*

$$R = (v_o \cos\theta)t$$



let, t_1 is the time interval to reach the *top* of the jump ($v_y = 0$)

$$v_y = v_{yo} - gt_1 \qquad 0 = 11\sin 20^o - (9.80)\,t_1,$$

$$t_1 = 0.384\ s$$

The *total* time in the air is: $t = 2t_1 = 0.768\ s$

Substituting into R gives

$$R = (11\cos 20)(0.768) = 7.94\ \text{m}$$

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(b) The maximum height reached

The vertical distance is given by

$$y = (v_o \sin\theta)t - \frac{1}{2}gt^2$$

The maximum height is reached when $t=t_1$, therefore

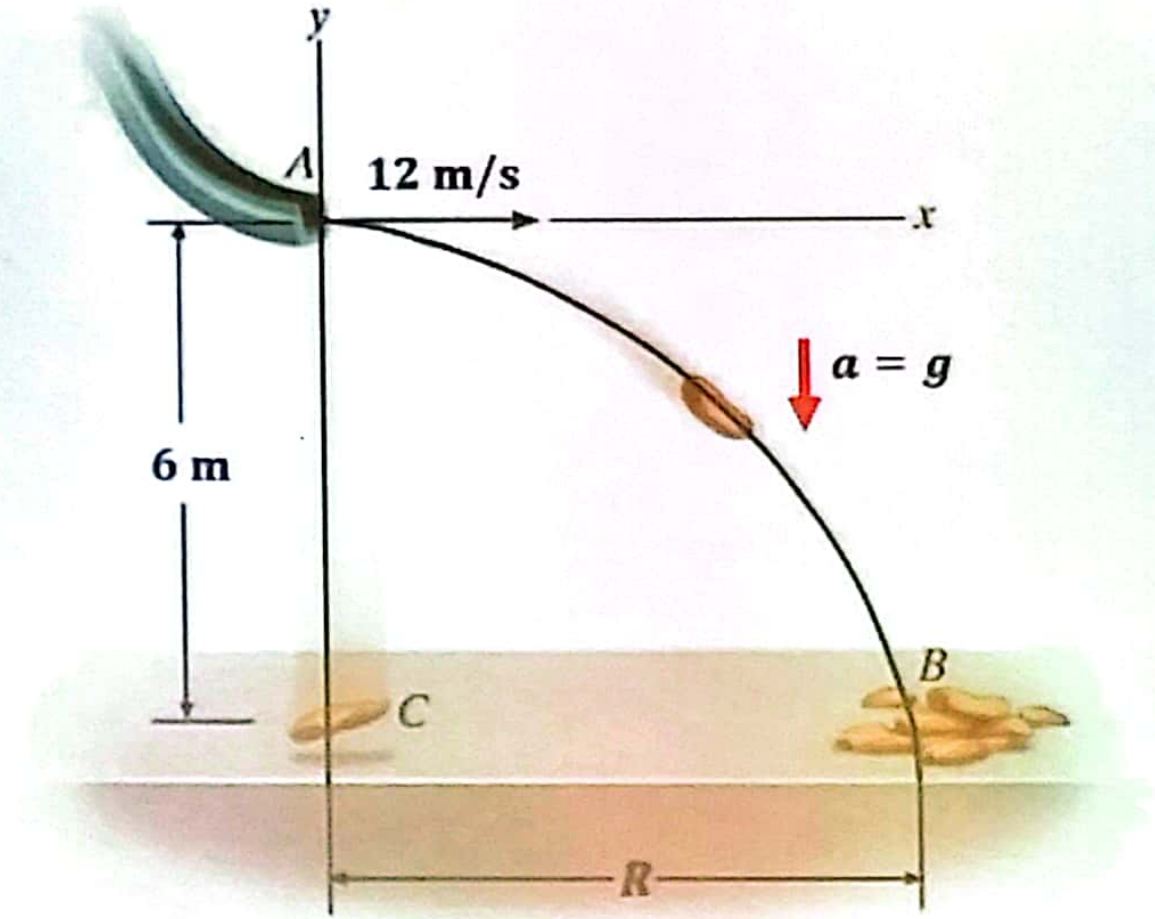
$$y_{max} = (11 \sin 20)(0.384) - \frac{1}{2}(9.8)(0.384)^2$$

$$= 0.722 \quad \text{m}$$

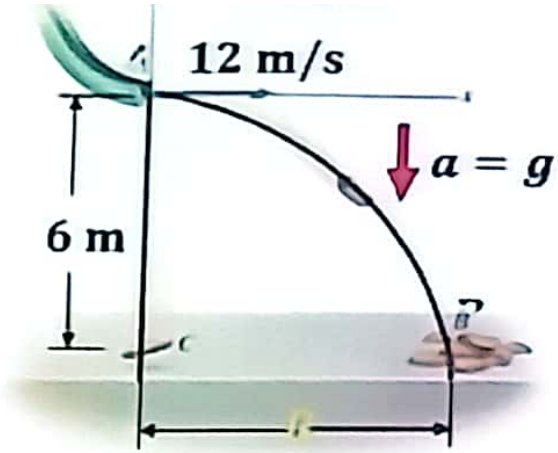
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Example

A sack slides off the ramp shown with a horizontal velocity of 12 m/s. If the height of the ramp is 6 m from the floor, determine the time needed for the sack to strike the floor and the range R where sacks begin to pile up.



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Vertical Motion.

المسافة الرأسية من A الى B معلومة ، ولذلك يمكن ايجاد الزمن t_{AB} مباشرة من العلاقة:

$$y_B = (v_A)_y t_{AB} + \frac{1}{2} a_c t^2{}_{AB}$$

$$-6 = 0 + \frac{1}{2}(-9.81) t^2{}_{AB}$$

$$\therefore \quad t_{AB} = 1.11\ s \qquad \textbf{Ans.}$$

Horizontal Motion

$$x_B = (v_A)_x t_{AB} \quad \rightarrow \quad R = 12\ (1.11) = 13.3\ \text{m}$$

Ans.

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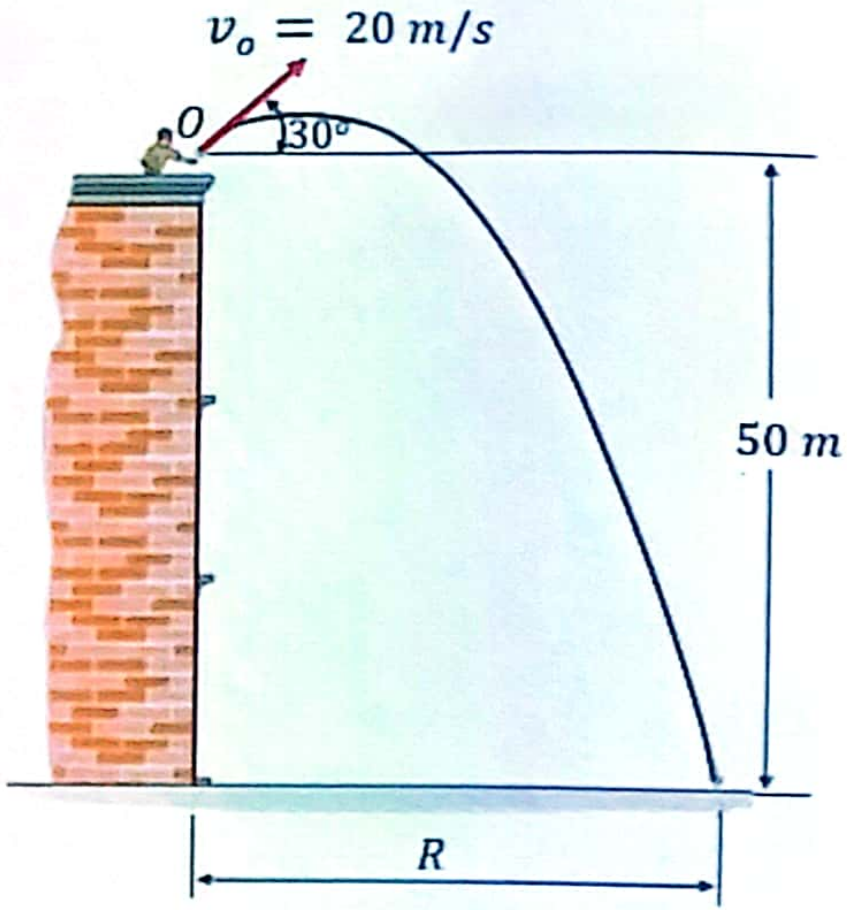
EXAMPLE

A stone is thrown off the top the building upward at an angle of 30° to the horizontal and with an initial speed of 20 m/s, as in Figure. The building is 50 m high,

(a) how long after being released does the stone strike the ground ?

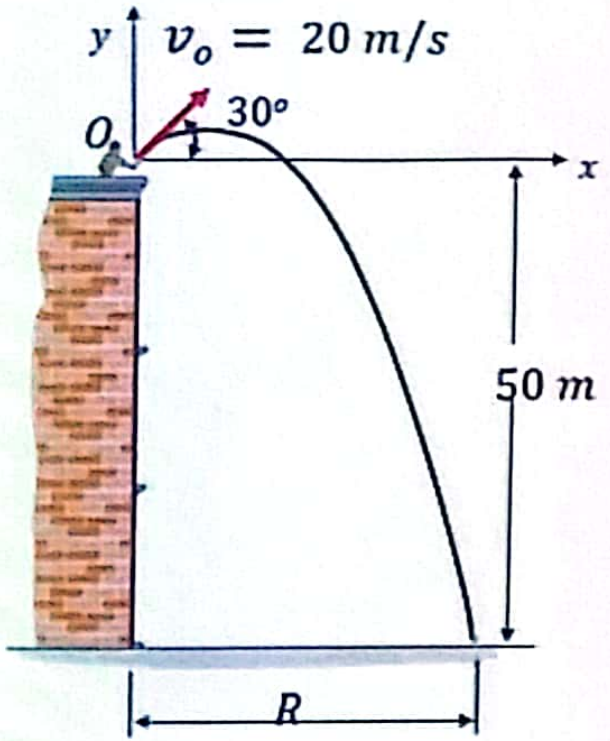
(b) With what speed and angle of impact does the stone land?

(c) Where does the stone strike the ground?



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SOLUTION



(a) Time of flight:

The initial x and y components of the velocity are

$$(v_0)_x = v_0 \cos\theta$$
$$= 20 \cos 30 = 17.3\ m/s$$

$$(v_0)_y = v_0 \sin\theta$$
$$= 20 \sin 30 = 10.0\ m/s$$

The vertical motion of the stone :

$$y = (v_0)_y t - \frac{1}{2} g t^2 \quad \rightarrow \quad -50 = 10t - \frac{1}{2}(9.8)t^2$$

The positive root of this equation gives: $t = 4.22\ s$

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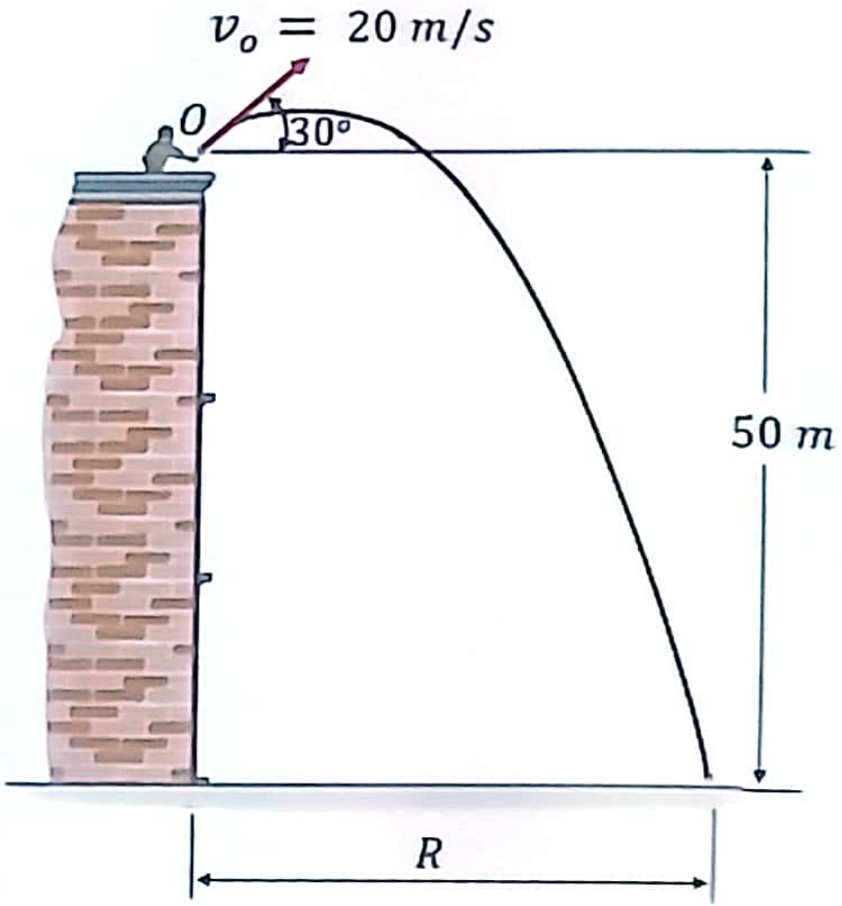
c) Range

$$R = (v_0)_x t$$

$$= 17.3(4.22)$$

$$= 73\ m$$

from the base of the building



A projectile is fired with the initial velocity of 90 m/s to hit a ground level target. Its maximum horizontal range will be ————————.

A. 81 m
B. 19.2 m
C. 800 m
D. 825.7 m

Answer: D

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In projectile motion a body moves with ——— ———.

A. horizontal component changing but vertical component of velocity constant.
B. constant vertical component of velocity
C. both changing horizontal and vertical components of velocity
D. constant horizontal component of velocity

Answer: D

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Small angle produces ————,

A. high trajectory
B. straight trajectory
C. steep trajectory
D. flat trajectory

Answer: C

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The total flight requires a time that is ———— ————.

A. twice the time necessary to reach the maximum height
B. four times the time necessary to reach the maximum height.
C. equal to the time necessary to reach the maximum height.
D. thrice the time necessary to reach the maximum height

Answer: A

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To attain maximum range, the projectile must be launched at an angle of ————.

A. 60°
B. 45°
C. 0°
D. 90°

Answer: B

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The range of the projectile depends on the square of the initial velocity and ————.

A. sine of thrice the projection angle θ
B. cot of twice the projection angle θ
C. sine of twice the projection angle θ
D. cosine of twice the project angle θ

Answer: C

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The motion of a projectile follows a ————————

.

A. vertical path.
B. spinning path.
C. curved path.
D. straight path

Answer: C

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A player making a long jump is an example of ——————.

A. Projectile motion.
B. Horizontal motion.
C. Spinning motion.
D. Rotatory motion.

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The motion on a curved path, when one component of velocity is constant and the other is variable is called ———

A. Projectile motion
B. circular motion
C. vibratory motion
D. spin motion

Answer: A

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If a projectile is launched at 45^0 with velocity 100 m/s, it hit the target. It will have double the range if its velocity is ——— ———.

A. 200 m/s
B. 173.2 m/s
C. 141.4 m/s
D. 400 m/s

Answer: 3

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